

Research Companion

User Manual - Version 0.5.8

Every feature of the tool, how to use it, and how to read its outputs honestly.

This manual explains everything Research Companion does, how to use it, and — just as important — how to read its outputs honestly. It assumes no prior knowledge of the tool.

1. What is Research Companion?

Research Companion is an open-source assistant for people writing and evaluating research papers. You give it papers — your own draft and the literature around it — and a team of specialized AI agents reads them, builds a shared knowledge graph, and answers the questions every author faces: **Are my references real? Is my contribution actually novel? Which papers support or threaten my draft, and where? Have I read everything my own bibliography cites?**

The one-sentence version: ***a team of AI helpers that reads your papers and checks its own answers before showing you.*** Most AI tools hand you fluent text and leave you to wonder what is real. Research Companion is built the other way around: every verdict carries evidence you can check, every quote is verified verbatim against the source text, and when something cannot be verified the tool says so on screen instead of hiding it.

It runs entirely on your machine (a local store in `~/.research-companion`), talks to an AI provider you configure (Anthropic or OpenAI) only when a task needs it, and costs pennies per paper. It ships with zero-key offline demos so you can see everything work before configuring any account.

Who it is for: researchers and students preparing a submission; anyone managing a growing pile of PDFs; reviewers and educators who want a working example of verification-first LLM system design.

2. Installation & first launch

```
pip install research-companion
research-companion lab serve
```

Your browser opens the **Research Lab** at `http://127.0.0.1:8765`. On first launch a four-step onboarding walks you in:

1. **Connect a model** — open Settings, paste an Anthropic or OpenAI API key. Keys are stored in a local `.env` file (never in JSON, never committed), masked everywhere in the UI, and take effect immediately — no restart.

2. **Add your draft** — the Add Papers dialog opens on the Upload tab; drag your PDF in and tick **"This is my draft ★"**.

3. **Ingest a folder** — point it at the directory where your PDFs already live and watch the knowledge graph grow in real time.

4. **Meet your suggestions** — the bell icon opens concrete, sectioned advice.

Zero-key demos (from a source checkout): `python examples/demo_offline.py`

runs the full review pipeline with no network; `python examples/demo_lab_offline.py` replays the Lab's event stream.

3. Core concepts

- **Research (workspace)** — one isolated project: its own papers, graph, draft, suggestions, gaps, timeline, saved views, and chat history. API keys and appearance settings are global; everything else is per-research.
- **Draft ★** — the one paper that is *yours*. Everything else is analyzed relative to it: alignments, suggestions, citation coverage, gaps.
- **Library** — the papers you have ingested. Each is split into sections, distilled into concepts/methods/datasets/claims/results, and merged into the cross-paper knowledge graph.
- **Verification** — the design principle. Citations are checked against live scholarly databases; evidence quotes are located verbatim in source text and badged verified/unverified; confidence scores are deterministic formulas with visible weights.

4. Researches (workspaces)

The top bar shows the active research as a dropdown (e.g. **"Main ■"**).

- **Switch** — pick another research from the dropdown; the whole Lab reloads into it. Other open tabs reload automatically.
- **All researches...** — a card-per-research overview screen (name, draft title, paper count, open suggestions, last activity) with **create**, **rename**, **archive**, and **delete** actions.
- **Delete** — the trash icon on any card (archived ones included) deletes that research and everything in it. The confirmation names the research and its paper count so you know exactly what you are removing. If you delete the research you are currently in, the Lab switches you to another one automatically (Main preferred) and reloads. Deletion is blocked with a clear message while a background job is running. **Archive** remains the non-destructive option — it hides a research without deleting anything.

- **CLI:** `research-companion workspace list|create <name>|use <id-or-name>;`
a single invocation can also be redirected with the `RESEARCH_COMPANION_WORKSPACE` environment variable.
- **Isolated per research:** papers, graph, draft pointer, suggestions, gaps, saved views, journey, conversations, event log. **Global:** API keys (`.env`), theme/accent/density, provider/model, retrieval knobs.
- **Upgrading from pre-0.4:** your existing store migrates automatically and losslessly into `workspaces/main/` on first run — atomic, resumable if interrupted, nothing re-ingested, `.env` untouched.

5. Adding papers

Click "+ Add papers" (top right). Three tabs:

- **Upload PDF** (default) — drag a PDF onto the dropzone or click to browse. The "**This is my draft ★**" checkbox uploads and marks your draft in one step. Duplicate uploads are recognized by content hash and answered gently ("Already in your library").
- **arXiv / URL / path** — type an arXiv ID (2312.12345), a DOI, a URL, or an absolute path to a PDF on disk.
- **Ingest folder** — a directory path; every PDF inside is discovered and processed. You can watch the pipeline live: sections parsed, entities extracted, graph nodes blooming, per-file progress in the dock.

Files that cannot be parsed (scanned/corrupt PDFs) become red failure cards with the exact reason and a one-click **Retry** — no silent failures.

6. The Library

Two views, toggled at the top right of the Library:

- **Grid** — cards with strength colors and stance chips.
- **List** — a sortable table: Title (your draft pinned first with ★), Year, **Status** (Queued → Processing → Ingested, or Failed with inline Retry), **Strength**, **Relation** to your draft (strengthens / challenges / alternative), and Added. Status pills update live during a folder ingest.

Strength is a deterministic triage score computed from four visible signals — extraction completeness (0.30), citation health (0.25), alignment with your draft (0.25), recency (0.20) — renormalized over whichever signals exist. Green = strong, yellow = moderate, orange = weak, gray = unscored. It is a convenience for triage, not a judgment of the paper's science.

Every card (grid) and every row (list) carries a **Remove** button, so you

never have to open the drawer just to delete a paper. Removing the paper marked as your draft ★ clears the draft flag automatically.

Click any row/card for the drawer: metadata, sections, alignment summary, **Set as draft / Unset draft** (marking a draft is no longer one-way), and Remove.

Paper metadata — automatic and manual

Every paper carries a **title**, **authors**, and **year**. These are no longer just the filename: when a paper is analysed the tool reads them straight from its text and fills them in automatically, and existing papers that predate this are backfilled once — any paper that has extracted text but no year is re-read the first time you open its workspace to recover its metadata. ****Scanned or image-only PDFs with no extractable text cannot be auto-filled**** — there is nothing to read — so you set their metadata by hand.

- **"Needs metadata" indicator.** Any paper still missing authors or a year wears an amber **"Needs metadata"** marker in the Library, and a banner tells you how many papers need attention so nothing quietly stays half-identified.
- **Edit it by hand.** Open the paper's drawer and click **Edit metadata** to reveal a form for the **title**, **authors**, and **year** (the year is constrained to 1900–2100). Fill in what is missing or correct what was extracted and **Save**; the indicator clears once the paper has both authors and a year.

Metadata is not cosmetic — two features depend on it. The **Timeline** (section 16) needs a year to place a paper at all, so papers only appear there once their year is set. **Citation coverage** (section 7) matches a cited reference to a paper you already own by **first-author surname + exact year** as well as by title, so accurate authors and year stop the tool from urging you to "Add" a paper that is already in your library.

7. Citation coverage — your bibliography as ground truth

When you set a draft, its References section is parsed (four splitting strategies with sanity checks) and every cited paper is matched against your library by arXiv ID, DOI, title, or **first-author surname + year**.

- **Automatic downloads** (default on): missing cited papers with a recognizable arXiv ID or DOI are downloaded and ingested with ****zero clicks**** the moment your draft or library changes. Title-only references

get one automatic resolution pass against CrossRef/OpenAlex — deliberately conservative, because a false match would download the wrong paper.

Settings → Citations → "Automatically download papers cited by your draft" is the off-switch (each download costs one model extraction call, ~\$0.02–0.10).

- **The Citations panel** lists every cited paper with its status:

In library ✓, *Downloading...*, *Add* ↓ (one-click, or "Add all"),

Not checked, *Unresolved* ? (could not be fetched — hover any row for the raw citation string).

- **The disclaimer banner** — whenever coverage is incomplete, every view shows a slim banner: *"Analysis covers N of M cited papers — Add missing."* Alignments, suggestions, and gaps are only as complete as the library they run against; the tool says so rather than letting partial coverage masquerade as a full picture. Collapsible per session; never disappears for good until coverage is complete.

- **Honest fallback:** if PDF text defeats bibliography parsing, the list falls back to the related-work extraction and says so in an amber note — labeled, never silent.

When automatic matching misses — a cited reference worded differently from a paper's real title, say — you can **link it by hand**, from either side:

- **From the Citations panel.** Each not-in-library citation row has a "Link..." control that opens an inline dropdown of your library papers (the ones still needing metadata are listed first, since they are the likeliest matches). Pick the paper and the citation is linked to it.
- **From the paper drawer.** Open a library paper and use **"This is a cited reference..."** to attach it to one of your draft's citations from the paper side.
- **What the titles and years mean.** The picker shows titles and years drawn from your **draft's citations**, not from the papers themselves — a disclaimer says so — so you are choosing on the citation's terms, not the library's.
- **What linking does.** The citation is marked **In library** and the mark is durable: it survives coverage recomputes and reverts only if you delete the paper it points to. Linking also **backfills the paper's year** from the citation (year only) when the paper had none, so a previously yearless paper appears on the **Timeline** (section 16) right away. Authors are left untouched — fill those in with **Edit metadata** (section 6) when you want them.

8. Citation placement

Citation coverage asks *whether* each cited paper is in your library.

Citation placement asks a different question: is each cited paper discussed in the draft section where it is **most relevant**? Open it from the **map-pin button in the top bar** (visible with a draft set). Each cited paper gets one of three verdicts:

- **Well-placed** — the paper is cited in the section it fits best.
- **Misplaced** — the paper would be more relevant in a different section than the one that cites it; the panel names the section it suggests.
- **Unknown** — placement could not be determined for that reference.

Limitation (v1). Placement only activates for ****numbered [n]** bibliographies** parsed from the draft's References section — the analysis needs the [n] markers to tie each in-text citation to a reference. Drafts that use author–year (e.g. (*Smith et al.*, 2021)) or any other non-numbered citation style show **"not applicable"** rather than a guess. There is no CLI command for placement; it is a Lab-only view.

9. The knowledge graph

Every paper becomes concepts, methods, datasets, claims, and results, merged across papers (five papers mentioning GraphRAG = one node, five connections).

- **Draft view** (default when a draft is set) — a deterministic layout with your draft at the center under a gold halo, its sections as an inner ring, and every library paper placed in labeled sectors by how it relates: STRENGTHENS / CHALLENGES / ALTERNATIVE / unaligned, with relation-colored edges (green / red / amber). Click a paper to expand its extracted entities. Same graph, same picture, every time.
- **Explore view** — the free-form physics layout for open-ended browsing, with kind filters and search.
- **Section subgraphs** — the left panel filters the graph to any section of your draft.
- **Saved views** — save the subgraph behind any Ask answer (or hand-picked nodes) as a named view; pin, rename, reload from the Graph sidebar.

10. Draft analysis (alignments)

Every library paper gets a verdict against *your* sections: which it **strengthens**, **challenges**, or offers an **alternative** to — each with

a relevance meter, a written rationale, and evidence quotes from the other paper **verified verbatim** against its text (✓ verified / unverified badge when the quote cannot be found). The Draft view shows your section tree with stance chips; "view in graph" jumps to that section's subgraph.

11. Reading papers and sections

The Lab has a built-in **reader** so the source text is always one click from any claim about it — you never have to leave the app to check what a paper actually says.

Ways to open it:

- **A draft section** — click a section in the draft view's section list, or a section in the **graph** section list, to open the reader on your draft's extracted text, scrolled to that section and highlighted.
- **A paper** — the graph node's **"Read paper"** action, the **Read** button on any library card or list row, or the same button in the paper drawer, opens that paper's full extracted text.
- **An evidence quote** — click a **verified evidence quote** on a draft alignment card and the cited paper opens with that exact quote highlighted.

Locating the quote is whitespace- and case-insensitive; if it cannot be located, the paper still opens with a notice rather than failing silently.

Inside the reader:

- A **section-navigation rail** lists the paper's sections; click any one to jump to it. The current section (or the located quote) is highlighted.
- **View original PDF** opens the stored PDF in a new browser tab.
- **Scanned PDFs** (and any paper with no extracted text) show a clear empty-state — *"No extracted text — use View original PDF"* — instead of a blank pane, so you always have a way to reach the source.

12. The six-lane review

research-companion review <paper-id> --serve --report out/ runs the full pre-submission review with a live dashboard:

Lane	What it does	What you get
citation	Resolves every bibliography entry against CrossRef → OpenAlex (arXiv IDs first)	verified / suspect / unverified per reference, with reasons
priorart	Retrieves related work via Semantic Scholar (OpenAlex fallback)	ranked closest prior papers

Lane	What it does	What you get
novelty	Extracts your claimed contributions with verbatim quotes; compares against prior art; verifies every quote	per-claim verdicts: novel / incremental / overlaps / anticipated
confidence	Deterministic score per claim from three signals (evidence verification 1.0, novelty confidence 0.8, citation health 0.6)	e.g. 0.72 ± 0.15 — the band widens when signals disagree
benchmark	Mines the graph and related work for evaluation benchmarks	suggested benchmarks you may be expected to report
rebuttal	(own command) grounds point-by-point reviewer responses in retrieved passages	replies with honesty badges

Lanes run concurrently with failure isolation: one failing lane degrades the report, never the run. -- fast runs only LLM-free lanes. Every run appends to a JSONL audit log.

13. Suggestions & revision tracking

Deterministic rules read your review, alignments, and gap map and produce concrete, sectioned advice: *discuss paper X as an alternative in related work, back this claim with evidence, fix this unverified citation*. Each suggestion has a severity, a rationale, and a source; they live in the bell panel with Dismiss / Discuss actions.

Revision tracking: when you upload a new version of your draft, conservative matchers detect what you incorporated and flip those suggestions to **addressed** automatically. Dismissals are sticky. The **Home** view shows your journey: version pills, open/addressed counts, a severity donut, and a "Do this next" card driven by simple rules.

14. Ask & search

Type a question in **Ask**; the tool ranks all paper sections and sends only the best slices to the model. Answers come back with numbered citation chips (hover for a mini-card, click through to the paper), a grounding strip ("Grounded in 6 sources across 2 papers"), and a warning panel listing any quoted span that could not be verified against the sources. Questions can be scoped to a single section of your draft.

Hybrid semantic search: add a free Hugging Face token in Settings and retrieval fuses BM25 with embedding similarity. Without the token, ranking is byte-identical to pure BM25 — a tested degradation contract.

15. The Companion (chat)

The floating chat button opens the Companion — a candid senior-colleague persona that can discuss any artifact: the review, a section's alignment, a specific suggestion (click **Discuss** on it), a gap, or a paper. Answers are grounded in retrieved sections with [S#] citations; quotes are verified verbatim; unverifiable spans are flagged, not hidden. Conversations persist on disk per research. With no review report yet, it degrades to a project overview — it always answers.

Each thread has a **Clear chat** button that wipes its history and deletes the conversation stored on disk — it works even if nothing has been persisted yet.

16. Timeline & gaps

The **Timeline** lays out concepts, methods, and datasets by year — when each first appeared and how it evolved. Toggle the **gap overlay**: amber diamonds are limitations papers admitted in their own limitation/future-work sections (quoted and verified verbatim); your draft is the blue diamond. Click a gap for the evidence quote and a Discuss button. Gaps your draft addresses are highlighted; open gaps relevant to your draft become suggestions.

17. Background activity

The Lab tells you what it is doing:

- **Topbar spinner** — appears whenever anything runs, with a live label: *"Downloading 1810.04805", "Checking references...", "3 tasks running"*.
- **Progress dock** (bottom) — one line per running task.
- **Downloading... chips** on citation rows being fetched right now.
- The coverage banner switches to in-flight text while downloads run.

A tab opened mid-work picks up running tasks immediately. Idle = invisible.

18. Settings reference

- **Model** — provider (Anthropic/OpenAI), model override, API keys (write-only fields, masked, stored in local `.env`, applied without restart).
- **Semantic search** — Hugging Face token for hybrid retrieval.
- **Appearance** — dark/light theme, three accents, comfortable/compact density.

- **Retrieval** — `k_sections` (default 6) and `char_budget` (default 8000) control how much context Ask/Companion retrieve.
- **Citations** — "Automatically download papers cited by your draft" (default on).
- **About** — replay the intro tour; links to docs.

19. CLI reference

Everything in the web UI has a command-line twin:

```
research-companion add <arXiv-id|DOI|URL|PDF...> # add papers (batch/--from-file supported)
research-companion build # run extraction, build the graph (cached)
research-companion cost-estimate # projected API cost before building
research-companion list | remove | stats | search # library management & keyword search
research-companion view # standalone interactive graph page
research-companion chat ["question"] # conversational Q&A (REPL without arg)
research-companion ask "question" # token-efficient, citation-checked QA
research-companion set-draft <paper-id> # anchor your draft
research-companion align <paper-id> # one alignment verdict
research-companion compare <A> <B> # side-by-side comparison
research-companion review <paper-id> [--serve] # the six-lane review (+ dashboard)
research-companion rebuttal <paper-id> <reviews> # grounded reviewer responses
research-companion refcheck <paper-id> # citation validator standalone
research-companion discover <topic> [--expand] # find new papers via Semantic Scholar
research-companion gaps | timeline # gap analysis / temporal view
research-companion export <format> # markdown, CSV, JSON, Obsidian vault
research-companion workspace list|create|use # manage researches
research-companion lab serve|ingest|failures # the web Lab
```

20. Data, privacy & costs

- **Everything is local.** The store lives at `~/research-companion/` (override with `RESEARCH_COMPANION_DIR`): `global .env` (keys), `settings.json`, `workspaces.json`, and one folder per research under `workspaces/` — plain JSON and PDFs you can inspect.
- **What leaves your machine:** paper text goes to the AI provider you configured, only when a task needs it (extraction, alignment, answers); bibliographic lookups query CrossRef/OpenAlex/arXiv/Semantic Scholar; optional embeddings go to the Hugging Face Inference API. Nothing else.
- **Costs, measured live:** extraction $\approx$ \$0.02 per paper (GPT-4o class); re-extracting a 16-paper library cost \$0.51; an alignment or a grounded

answer is one model call. Everything cached is free to re-run.

21. Reading outputs honestly

- **"Verified" means the quote exists** in the source text — it does *not* mean the claim is true. A real paper can be cited for a false claim; quote verification cannot detect that, and the tool never claims it can.
- **Quote verification is paraphrase-blind.** A fabricated paraphrase passes unflagged as long as nothing is quoted.
- **Novelty verdicts are relative to what retrieval surfaced.** If search missed the killer prior paper, the verdict cannot know about it. A pilot against human reviewers (n=10, published as an honest null) found the system identifies genuinely novel papers but under-penalizes weak ones — treat novelty as a structured second opinion, not a referee.
- **Scores are honest guesses.** Confidence weights (1.0/0.8/0.6) and strength weights (0.30/0.25/0.25/0.20) were chosen by judgment, not tuned on held-out data — they are printed on screen precisely so you can disagree.
- **Coverage caveats are load-bearing.** The "Analysis covers N of M cited papers" banner and the amber related-work-fallback note exist because partial context should never masquerade as a full picture.

22. Troubleshooting & FAQ

- **A PDF failed to ingest** — usually scanned (no text layer) or corrupt. The failure card shows the exact reason; Retry after fixing the file. OCR is not yet supported.
- **"Bibliography could not be parsed" (amber note)** — two-column PDF text sometimes defeats the splitter; coverage falls back to the related-work extraction, clearly labeled. Hover rows to see raw strings.
- **"Connect a model" shows although I set a key** — keys entered in Settings apply instantly; keys set via environment variables are also detected. Check Settings → Model shows the provider you intend.
- **A cited paper stays "Unresolved"** — the conservative resolver refused to guess. Add it manually via the arXiv/DOI tab; precision beats a wrong download.
- **Where did my pre-0.4 store go?** — migrated automatically into `~/.research-companion/workspaces/main/`; nothing was lost.
- **Switching research reloads other tabs** — intentional; a tab must never show one research's data against another's store.

Roadmap (planned, not yet shipped)

An MCP trust server ("a local, bounded, key-free trust server for citation verification, quote grounding, citation coverage, and library search"), biomedical/CS metadata connectors (PubMed, Europe PMC, DBLP), and venue-checklist skill packs. See docs/superpowers/specs/ for the design.

Companion document: docs/RESEARCH_COMPANION_GUIDE.pdf — the guide & critical analysis, including measured evaluations and known failure modes.